CHAPTER 11

Mechanisms and Models of Nuclear Reactions

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The variety and complexity of nuclear reactions make this a fascinating area of research quite apart from the practical value of understanding fusion and fission. From studies of such properties as the relative amounts of formation of various competing products, the variation of the yields of these with bombarding energy, the directional characteristics and kinetic energies of the products, etc., we may formulate models of nuclear reaction mechanisms. Such models lead to systematics for nuclear reactions and make possible predictions of reactions not yet investigated. As such, the nuclear reactions can be simulated in fashons very similar to those used for thermodynamic modelling of chemical reactions where the yield of different species are governed by stability constants which in turn may be differently affected by, e.g. changes in temperature etc.

11.1. THE REACTION CROSS-SECTION

The probability for a nuclear reaction to occur is expressed in terms of the *reaction cross-section*. The geometric cross-section that a nucleus presents to a beam of particles is πr^2. If

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we use 6×10^{-15} m as an average value for the nuclear radius, the value of πr^2 becomes $3.14\ (6 \times 10^{-15})^2 \approx 10^{-28}$ m^2. This average geometric cross-section of nuclei is reflected in the unit of reaction probability which is the *barn,* b, where $1\ \text{b} = 10^{-28}$ m^2. The name barn has a historic background – cross sections of $\sim 10^{-28}$ m^2 were said to be "big as a barn".

Consider the bombardment of a target containing N_v atoms per m^3 by a homogeneous flux φ_o of particles (Fig. 11.1). The flux is expressed in units of particles m^{-2} s^{-1}. The target atoms N_v refer only to the atoms of the species involved in the nuclear reaction. If a Li-Al alloy is bombarded to induce reactions with the lithium, N_v is the number of lithium atoms per m^3 in the alloy, not the total of lithium and aluminum atoms. The change in the flux, $d\varphi = \varphi - \varphi_o$, may be infinitesimal as the particles pass through a thin section of target thickness dx. This change depends on the number of target atoms per unit area (i.e. $N_v dx$), the flux ($\varphi_o \approx \varphi$), and the reaction cross-section σ.

$$-d\varphi = \varphi\, \sigma\, N_v dx \tag{11.1}$$

The negative sign indicates that the flux decreases upon passing through the target due to reaction of the particles with the target atoms: thus $-d\varphi$ is the number of reactions. Integration gives:

$$\varphi = \varphi_o e^{-\sigma N_v x} \tag{11.2}$$

where φ_o is the projectile flux striking the target surface. For targets which have a surface area of S (m^2) exposed to the beam, for the irradiation time t, the total number of nuclear reactions ΔN is:

$$\Delta N = (\varphi_o - \varphi) S\, t \equiv \varphi_o S\, t\left(1 - e^{-\sigma N_v x}\right) \tag{11.3}$$

For a thin target in which the flux is not decreased appreciably upon passage through the target, i.e. $\sigma\ N_v\ x \ll 1$ and hence $e^{-\sigma N_v x} \approx 1 - \sigma\ N_v\ x$, (11.3) can be reduced to:

$$\Delta N = \varphi_o S\, t\, \sigma\, N_v x = \varphi_o \sigma\, t\, N_v V = \varphi_o \sigma\, t\, N_t \quad \text{(thin target)} \tag{11.4}$$

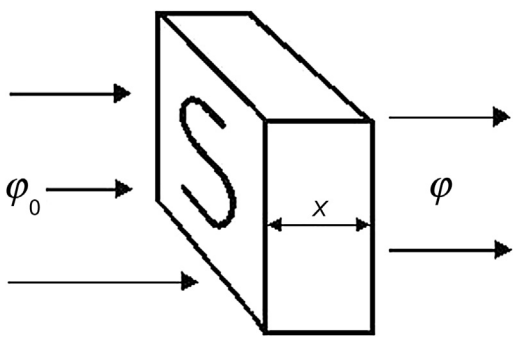


Figure 11.1 Reduction of particle flux by absorption in a target.

where $V = S\,x$ is the target volume, and $N_t = N_v\,V$ is the number of target atoms. Notice that as a result of the product $S\,x$, which equals the volume of the target, the relationship on the right of (11.4) is independent of the geometry of the target and involves only the total number of atoms in it.

Equation (11.4) can be used only when particle fluxes are homogeneous over the whole irradiated sample. In nuclear reactors, where the area of the sample is much smaller than the area of the flux, it is convenient to express the flux in terms of neutrons $m^{-2}\,s^{-1}$ and the target in terms of total number of atoms, as above.

By contrast, in an accelerator the target surface is often larger than the cross-section of the ion beam, and (11.4) cannot be used without modification. For accelerators the beam intensity (i.e. particle current) I_o (particles s^{-1}) is given by (16.3); 1 A (ampere) correspond to 6.24×10^{18} times the charge of the particle per second. Hence we have $½ \times 6.24 \times 10^{18}$ particles per second for a 1 A He^{2+} beam. In (11.4) $\varphi_o\,S$ must be substituted by $6.24 \times 10^{18}\,i\,z^{-1}$, where i (A) is the electric current and z is the number of electron charges.

In (11.4) $N_v\,x$ has the dimensions of atoms m^{-2}. This is a useful quantity in many calculations and for a pure elemental target is equal to

$$N_v x \;=\; 1000\,\mathbf{N}_A \rho\; x\; \gamma_i / M \tag{11.5}$$

where ρ is the density of the target (kg m^{-3}), M the atomic weight (in gram/mole) of the element, x the target thickness in m, and γ_i the isotopic fraction of reactive atoms of kind i in the target.

As an example of the use of these equations, consider the irradiation of a gold foil by thermal neutrons. Assume the foil is 0.3 mm thick with an area of 5 cm^2 and the flux is 10^{17} n $m^{-2}\,s^{-1}$. The density of gold is 19.3 g cm^{-3} while the cross-section for the capture of thermal neutrons by ^{197}Au is 99 b. Transferring these units into SI units and introducing them into (11.3) for an irradiation time of 10 min yields a value of 4.6×10^{15} for the number of ^{198}Au nuclei formed. If the thin target equation is used (11.4), the value of 5.0×10^{15} nuclei of ^{198}Au is obtained. If the same gold foil is bombarded in a cyclotron with a beam of protons of 1 μA when the cross-section is 1 b and $t = 10$ min, the number of reactions is 6.6×10^{12}.

Frequently, the irradiated target consists of more than one nuclide which can capture bombarding particles to undergo reaction. The *macroscopic* cross-section, which refers to the total decrease in the bombarding particle flux, reflects the absorption of particles by the different nuclides in proportion to their abundance in the target as well as to their individual reaction cross-sections. Assuming that the target as a whole contains N_v atoms m^{-3} with individual abundances γ_1, γ_2, etc., for

nuclides 1, 2, etc., the individual cross-sections are σ_1, σ_2, etc. The macroscopic cross-section $\hat{\sum}(m^{-1})$ is

$$\hat{\sum} = N_v \sum_1^n \gamma_i \sigma_i \tag{11.6}$$

For a target which is x m thick one obtains

$$\varphi = \varphi_o e^{-\hat{\sum} x} \tag{11.7}$$

The value $\hat{\sum}^{-1}$ is the average distance a projectile travels between successive collisions with the target atoms (the *mean free path*).

11.2. PARTIAL REACTION CROSS-SECTIONS

The irradiation of a target may lead to the formation of a number of different products. For example, the irradiation of ^{63}Cu with protons can produce the nuclides ^{62}Zn, ^{63}Zn, ^{62}Cu, all of which are radioactive:

$$^{63}\text{Cu} + {}^{1}\text{H} \rightarrow \begin{cases} (\text{p,n}) & {}^{63}\text{Zn} + \text{n} & (11.8a) \\ (\text{p,2n}) & {}^{62}\text{Zn} + 2\text{n} & (11.8b) \\ (\text{p,pn}) & {}^{62}\text{Cu} + {}^{1}\text{H} + \text{n} & (11.8c) \end{cases}$$

The formation probability of each product corresponds to a *partial reaction cross-section*. The total reaction cross-section σ_{tot} is the sum of all the partial cross-sections and measures the probability that the projectile causes a nuclear reaction independent of the products formed. Thus, the decrease in intensity of the particle flux is proportional to σ_{tot}. The amount of an individual product formed is proportional to σ_i, where σ_i corresponds to the partial reaction cross-section for the formation of the i:th product.

Some partial cross-sections have their own names such as the *scattering cross-section* for elastic and inelastic scattering (σ_{scat}), the *activation cross-section* (σ_{act}) for the formation of radioactive products, the *fission cross-section* (σ_f) for fission processes, and *adsorption* or *capture cross-sections* (σ_{abs} or σ_{capt}) for the absorption or capture of particles. If all of these processes take place, one obtains (with caution to avoid overlapping reactions)

$$\sigma_{tot} = \sigma_{scat} + \sigma_{act} + \sigma_f, \text{ etc} \tag{11.9}$$

In the irradiation of ^{235}U with thermal neutrons, σ_{scat} is about 10 b, σ_{act} (for forming ^{236}U) is approximately 107 b, and σ_f is 582 b.

The reaction cross-section depends on the projectile energy as shown in Figure 11.2. The curves obtained for the partial reaction cross-section as a function of projectile energy are known as *excitation functions* or *excitation curves*.